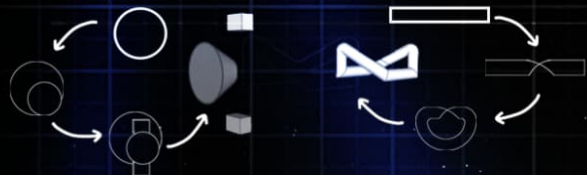


SOLACE

A HOME THAT PROVIDES COMFORT, PEACE, AND EMOTIONAL WELL-BEING DESPITE THE HARSHNESS OF SPACE.



FORM DEVELOPMENT

STORY

Year: 2198

For centuries, humanity pushed Earth beyond its ecological limits. Rising temperatures, ocean acidification, biodiversity collapse, and resource wars transformed once-thriving regions into uninhabitable landscapes. Although Earth still exists, it can no longer sustain a healthy human civilisation.

Rather than abandoning humanity, an international coalition launched the Infinity Habitat Initiative—a fleet of self-sustaining space homes designed to preserve life, culture, and knowledge while scientists worked to restore the planet over several generations.

The family of four is not escaping Earth; they are custodians of its future.



LEVEL 3 PLAN



LEVEL 2 PLAN



LEVEL 1 PLAN

A SPACE HOME DESIGNED NOT ONLY TO SUSTAIN LIFE, BUT TO SUPPORT THE EMOTIONS, RELATIONSHIPS, AND EVERYDAY EXPERIENCES THAT MAKE LIFE



SITE

Low Earth Orbit (LEO)

Orbital Altitude: 500–600 km above Earth

Orbit Type: Circular Low Earth Orbit

Occupants: Family of 4

Purpose: Long-term residential living, research, recreation, and observation



MATERIAL

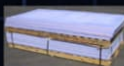
Carbon Fiber Reinforced Composite used in Primary structural frame
Lightweight and extremely strong making it ideal for the main structure while minimizing launch mass.



Aluminum-Lithium Alloy in External shell and pressure hull
Aerospace-grade alloy that provides high strength, corrosion resistance, and durability under extreme space conditions.



Transparent Aluminum (ALON) / Multi-Layer Smart Glass for Observatory deck and panoramic windows
A high-strength transparent material combined with electrochromic technology that automatically darkens when exposed to intense sunlight or solar radiation and becomes clear for unobstructed views of Earth and deep space.



Polyethylene Radiation Shielding in Interior wall layers
Hydrogen-rich material that effectively reduces cosmic radiation, creating a safer living environment for long-duration missions.



ETFE Cushions in Greenhouse and plantation dome
A lightweight, transparent membrane that allows maximum light transmission for plant growth while resisting harsh space conditions.



The control room and navigation center

observatory and anti-gravity zone



Uses of different zones of-mobius strip form in the space home



Escape pods transition meditation space

experimental zone and additional plantation with experimentation

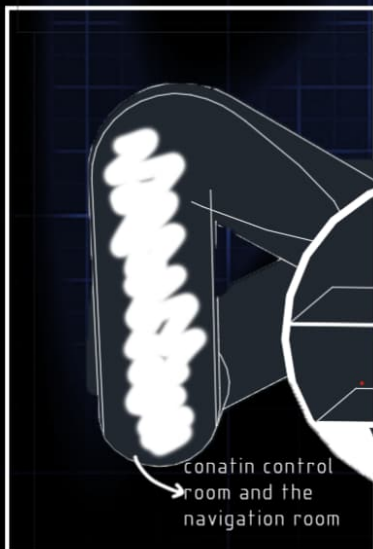
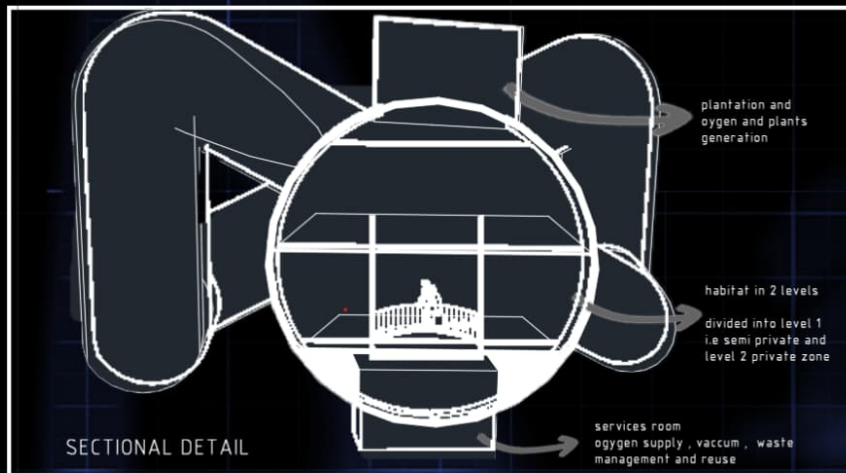
SITE ANALYSIS

The home remains in continuous orbit around Earth, maintaining visual and emotional connection with the planet while enabling efficient communication and future transportation.

The site offers uninterrupted panoramic views of Earth, Sunrise & Sunset, Deep Space, Moon, Stars, Aurora (depending on orbit). These views influence the placement of observation decks and shared family spaces, ideal conditions for Scientific experiments, Space observation, External maintenance.

Exposure to intense solar and cosmic radiation requires radiation-shielded living quarters, Water-wall protection, Emergency solar storm shelter, thermal Conditions

Continuous exposure to sunlight makes solar energy the primary renewable power source, supported by onboard energy storage systems.



S
O
L
A
C
E

